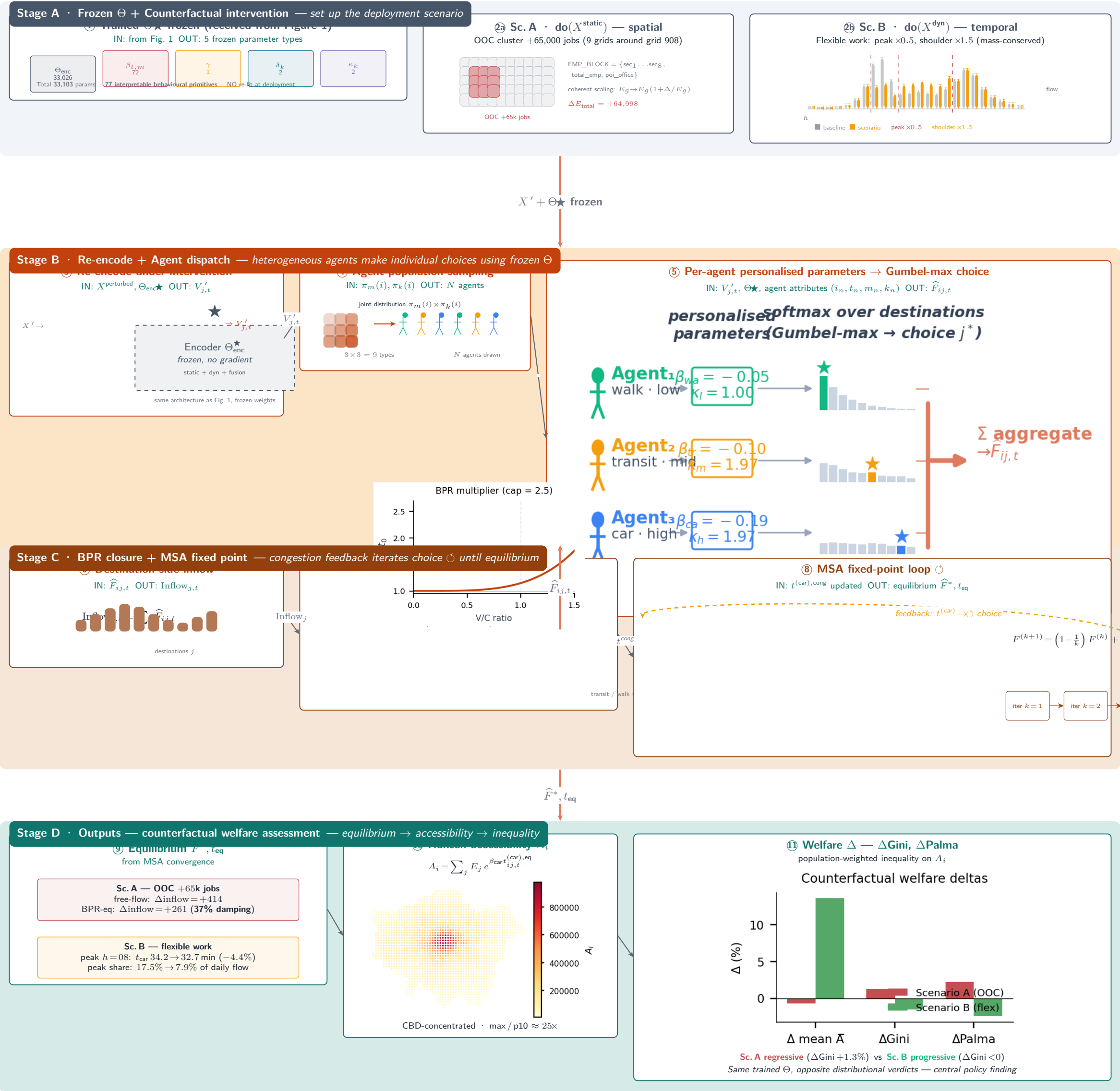


Figure 2 — ABM Forward Simulation Pipeline  
counterfactual deployment with frozen trained  $\Theta$  + destination-side BPR closure



Stage A receives the 5 trained parameter types from Figure 1. Stage B re-encodes  $X'$  with frozen weights, samples  $N$  heterogeneous agents from  $(\pi_m, \pi_k)$ , and dispatches each agent's individual Gumbel-max choice using their personal  $(\beta_{t,m,n}, \kappa_{k,n}, \delta_{k_n})$ . Stage C closes the destination-side congestion feedback loop until MSA convergence. Stage D reports equilibrium flow, Hansen accessibility, and population-weighted inequality measures.